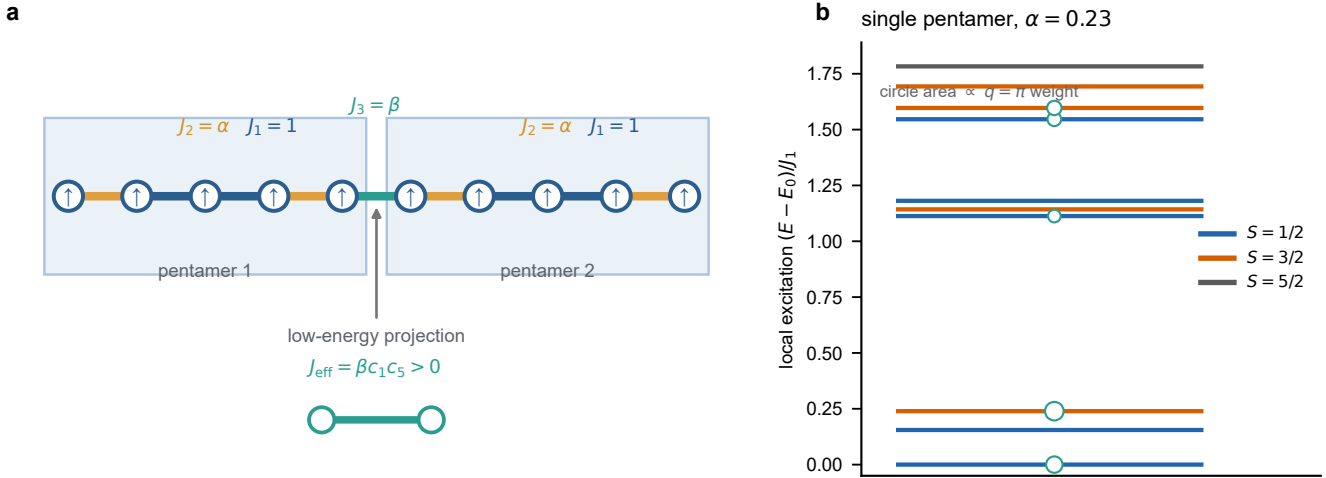
a
$J_3 = \beta$
$J_2 = \alpha$ $J_1 = 1$
$J_2 = \alpha$ $J_1 = 1$
pentamer 1
pentamer 2
low-energy projection
$J_{\text{eff}} = \beta c_1 c_5 > 0$
b
single pentamer, $\alpha = 0.23$
circle area $\propto$ $q = \pi$ weight
local excitation $(E - E_0)/J_1$
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
$S = 1/2$
$S = 3/2$
$S = 5/2$